INTRODUCTION

In this analysis, we will explore a traffic accidents in the Netherlands. The dataset contains information about the nature, cause, and outcomes of accidents, inorder to evaluate claims filed by the insured and also identify preventative measures to improve road safety.
The dataset includes key variables such as the location of accidents (latitude, longitude, and county names), weather conditions (precipitation, wind speed, temperature), offender details (e.g., hit-and-run incidents, uninsured drivers), casualty information (e.g., age, type of injury), and accident characteristics (e.g., road situation, type of accident).

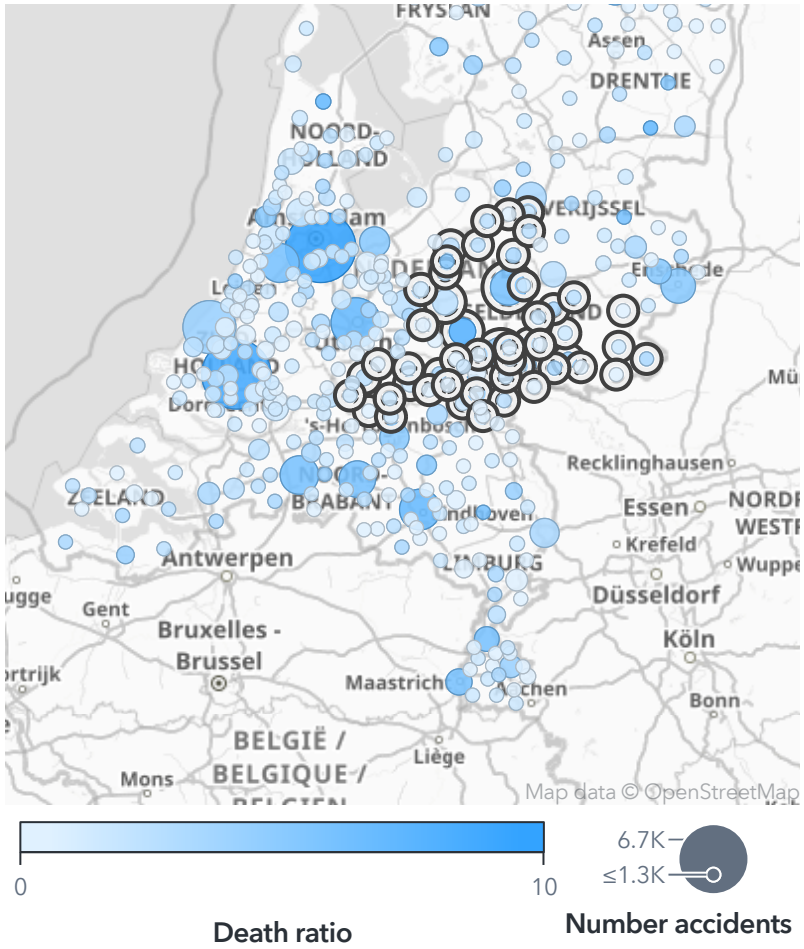
These variables provide a comprehensive overview of the factors contributing to traffic accidents in the Netherlands.
Through this analysis, we aim to answer key questions about traffic safety:

Where do accidents happen?
When do accidents happen?
Who are the offenders?
Who are the casualties?
What are the type of accidents?

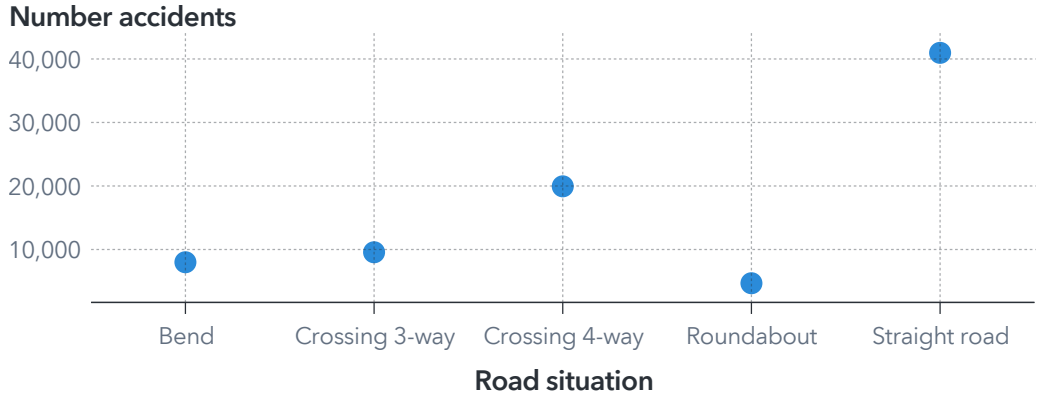
By uncovering patterns and trends in the data, we can identify factors that increase the frequency and severity of accidents. The results of this analysis will provide actionable insights to guide policymakers, road safety departments, and insurance providers in implementing more effective strategies to prevent accidents and improve road safety.

1. WHERE?

Death ratio by Province sized by Number accidents



Number accidents of Road situation



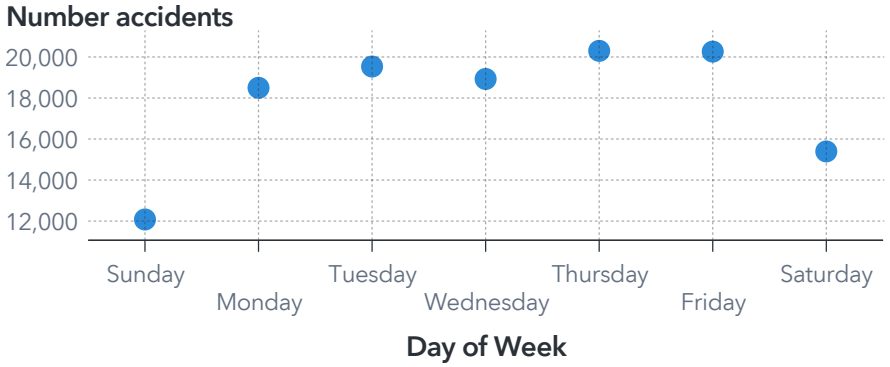
A1.1

County name	Number accidents	Number casualties	Death ratio ▼
Amsterdam	6,738	1519	10.00
Rotterdam	6,528	1479	9.00
Ede	957	92	8.00
Hollands Kroon	315	68	7.00
Coevorden	219	36	7.00
Utrecht	3,375	677	7.00
Eindhoven	2,267	226	7.00
Haarlemmermeer	2,068	338	6.00
Twenterand	202	27	6.00

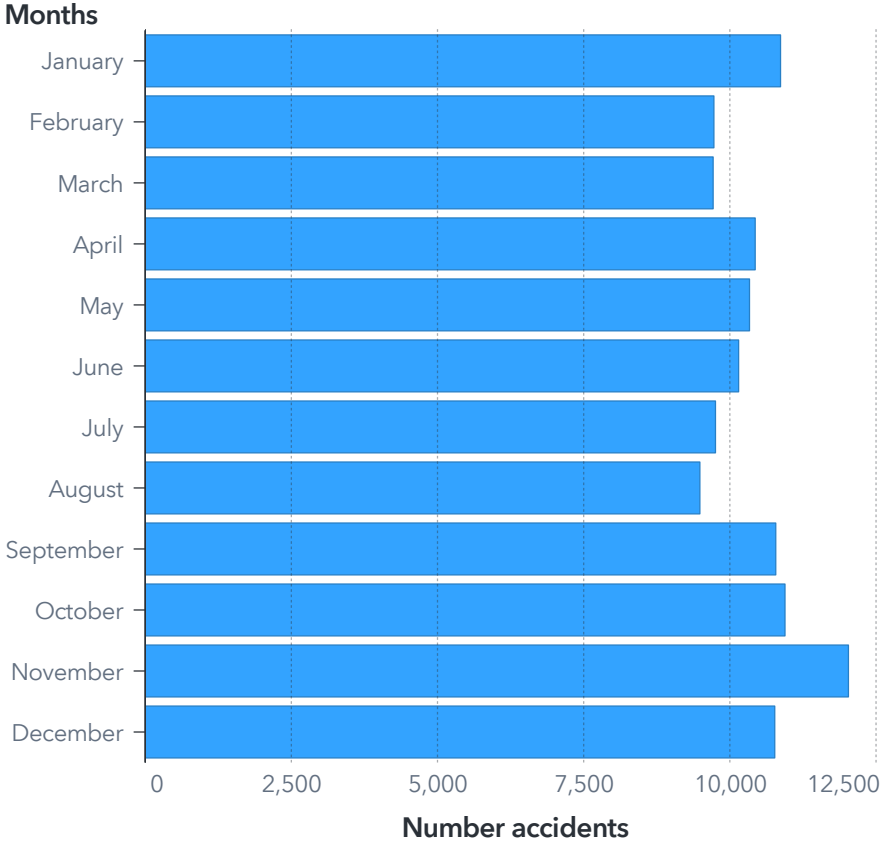
In the geoplot, we can clearly see that accidents are concentrated in specific provinces, with a higher death ratio observed in areas with more accidents. Larger bubble sizes indicate regions with more accidents.
In the plot of road situations, we can see that straight roads have the highest number of accidents, followed by crossings and bends, while roundabouts have the lowest.
Amsterdam stands out with the highest death ratio and number of casualties compared to other counties.

When do Accidents happen?

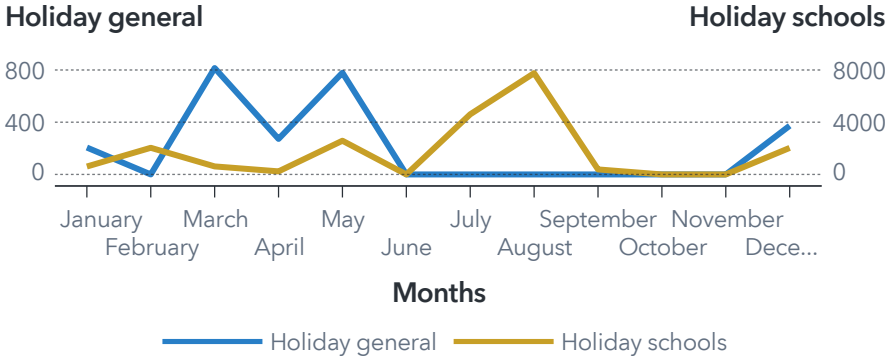
Number accidents of Day of Week



Number accidents of Months



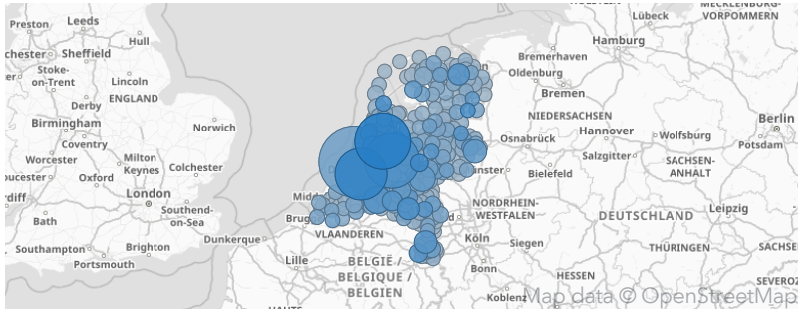
Holiday general, Holiday schools by Months



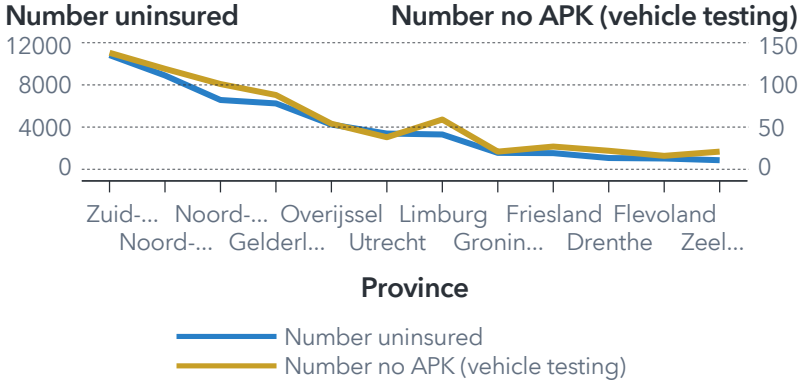
In the point plot for day of the week, we can see that accidents peak on Fridays and are lowest on Sundays.
In the bar chart for accidents by month, we can see that November has the highest number of accidents, while February and March has the lowest.
In the dual-line graph for holidays, we can see that holiday schools have a significant peak during the summer months (July-August), whereas general holidays have scattered spikes.

Who are the offenders?

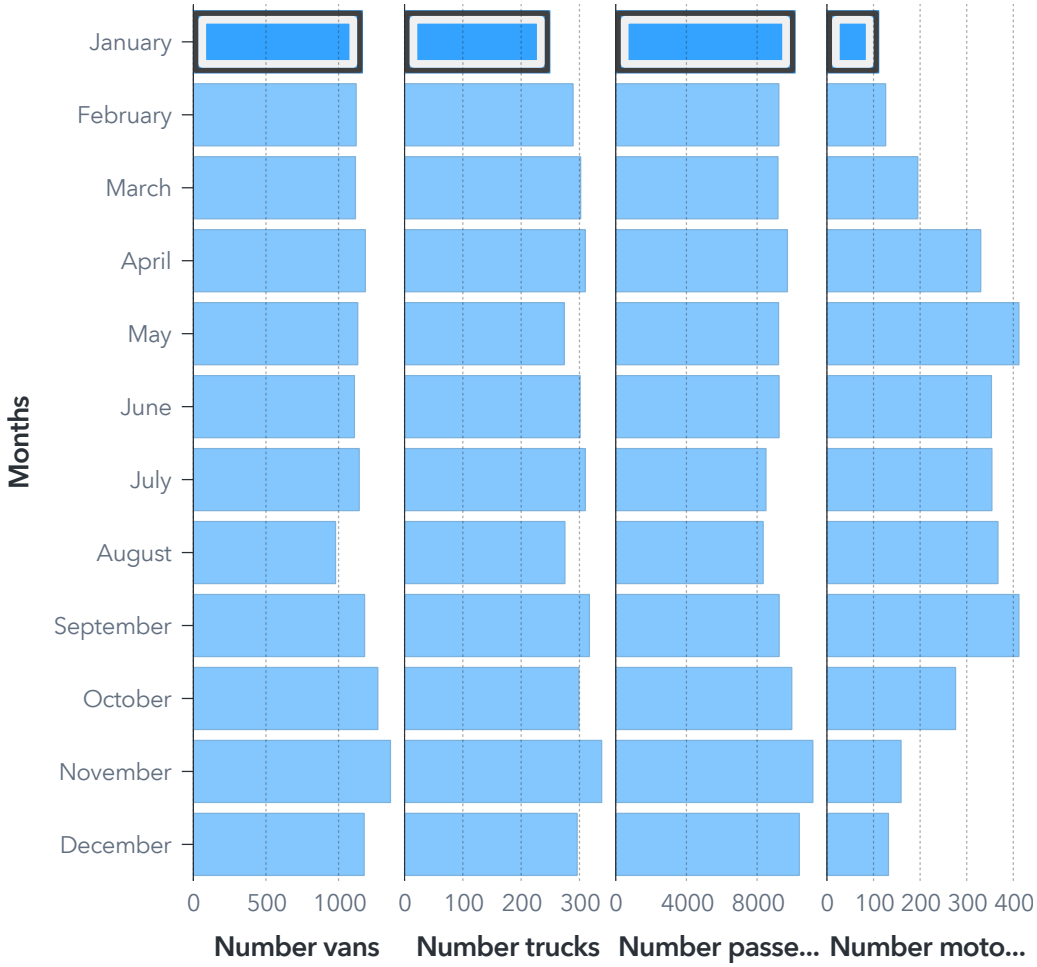
Death ratio by Province sized by Number hit-and-run



Number uninsured, Number no APK (vehicle testing) by Province



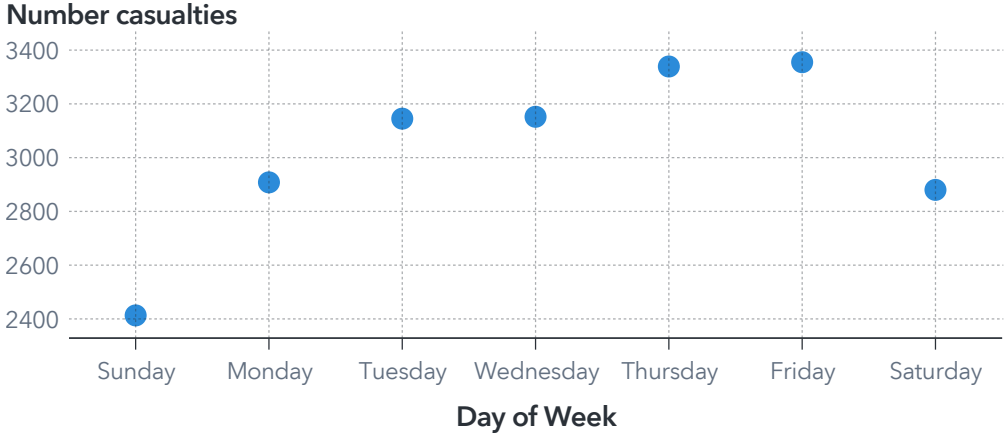
What were the offenders driving?



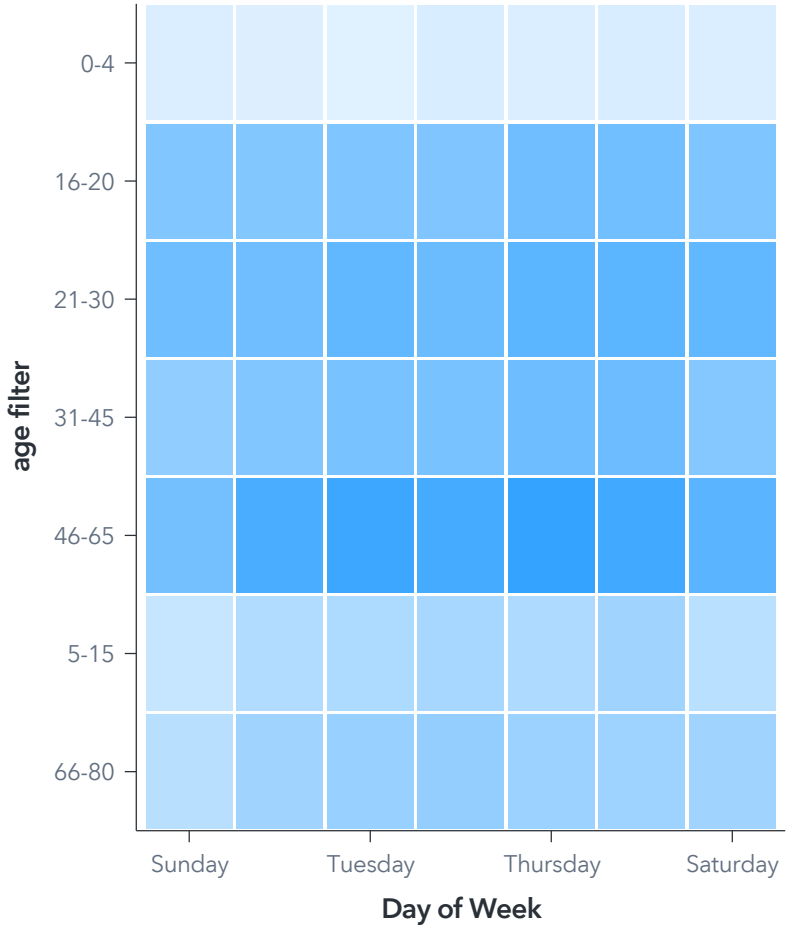
In the geoplot of death ratio by province sized by the number of hit-and-run incidents, we can see that certain provinces with higher hit-and-run cases correspond to higher death ratios.
In the line chart comparing the number of uninsured vehicles and vehicles without APK (vehicle testing) by province, we can see that Zuid-Holland and Noord-Brabant lead in both categories.

Who are the casualties?

Number casualties by Day of Week

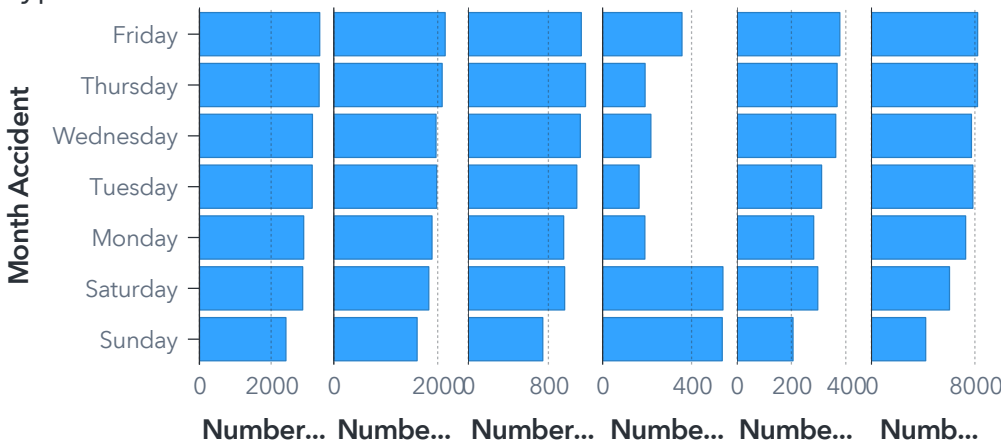


Age group of casualties



A2.1

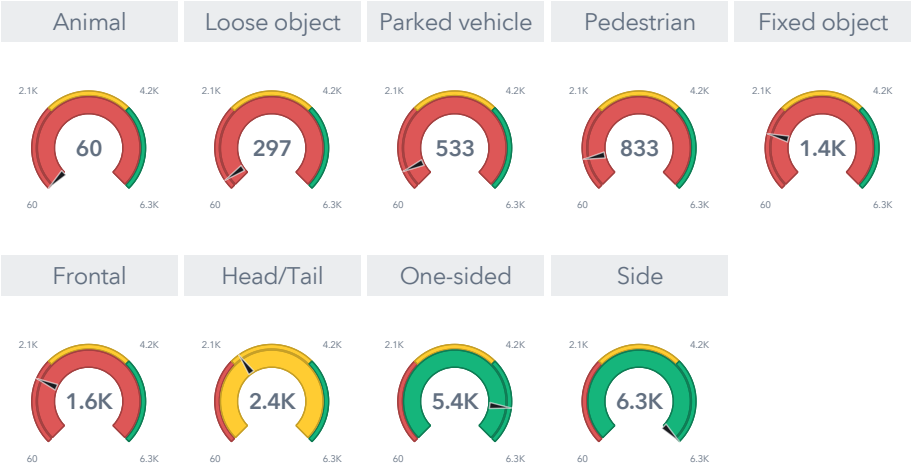
Type of casualties



In the scatter plot for casualties by the day of the week, we can see that Fridays have the highest number of casualties, while Sundays have the lowest. In the heatmap for the age group of casualties by the day of the week, we can see that the 46-65 age group consistently shows the highest density of casualties across all days.

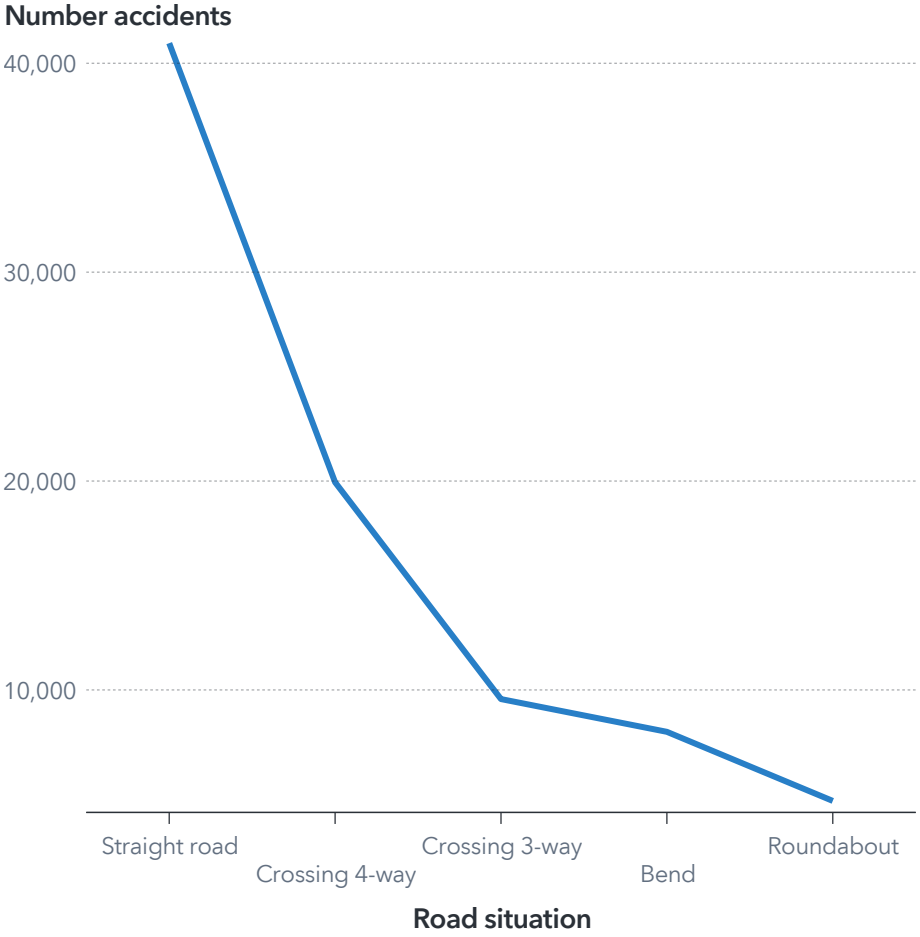
What are the types of accidents?

Number of casualties per type



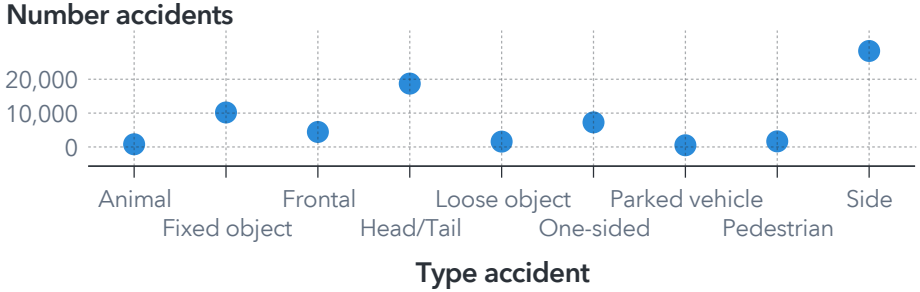
A3.1

Number accidents per Road situation



A3.2

Number accidents per accident



A3.3

In the gauge chart, we can see that side and one-sided accidents lead to the highest casualties, while animal-related accidents show the least.
In the scatter plot for accident types, we can see that side accidents are the most frequent, followed by head/tail.
In the line chart for road situations, we can see that straight roads account for the most accidents, whereas roundabouts have the least.

CONCLUSION

In this analysis, we can see that accidents are heavily concentrated in urban areas, with Amsterdam recording the highest death ratio and casualties. Straight roads are the most accident-prone, while roundabouts remain the safest.
Accidents are most frequent on Fridays and in November, while Sundays and February experience the fewest. Summer months (July-August) see a significant rise in accidents during school holidays, whereas general holidays show scattered peaks throughout the year.
We can see that Zuid-Holland and Noord-Brabant lead in uninsured vehicles and hit-and-run cases, making them high-risk provinces. Passenger cars dominate the vehicle types involved in accidents, followed by vans and trucks.
In terms of casualties, Fridays see the highest numbers, and individuals aged 46-65 are consistently the most affected group. Hospitalization is the most common casualty type, with Fridays showing the highest counts across all days.
Finally, we can see that side and one-sided accidents are the most frequent and result in the most casualties. Straight roads contribute the most to these accidents, while roundabouts remain significantly safer.

Recommendations

We can see that increasing safety measures on straight roads, such as speed monitoring and improved signage, could help reduce accidents. Targeted awareness campaigns on Fridays and in November can focus on high-risk behaviors. Stricter regulations and penalties for uninsured vehicles and improved vehicle testing compliance in high-risk provinces like Zuid-Holland and Noord-Brabant can make a significant impact. Enhancing road safety education, especially for passenger car drivers and older age groups (46-65), can also help reduce casualties.

Limitations

We can see that the data may not include unreported accidents, which could affect the overall accuracy. Geographical and seasonal variations in reporting could also influence the trends observed. Additionally, casualty data does not account for the severity or long-term outcomes of accidents.

Appendix

A1.1 Number accidents of Road situation

Filters: Road situation NOT MISSING

A2.1 Age group of casualties

Filters: age filter IN { '0-4'; '16-20'; '21-30'; '31-45'; '46-65'; '5-15'; '66-80' }

A3.1 Number of casualties per type

Filters: Type accident IN { 'Animal'; 'Fixed object'; 'Frontal'; 'Head/Tail'; 'Loose object'; 'One-sided'; 'Parked vehicle'; 'Pedestrian'; 'Side' }

Display Rules:

Number casualties

- $60 \leq x < 2142$
- $2142 \leq x < 4224$
- $4224 \leq x \leq 6306$

A3.2 Number accidents per Road situation

Filters: Road situation NOT MISSING

A3.3 Number accidents per accident

Filters: Type accident IN { 'Animal'; 'Fixed object'; 'Frontal'; 'Head/Tail'; 'Loose object'; 'One-sided'; 'Parked vehicle'; 'Pedestrian'; 'Side' }